Modularity-Based Algorithms for Community Detection

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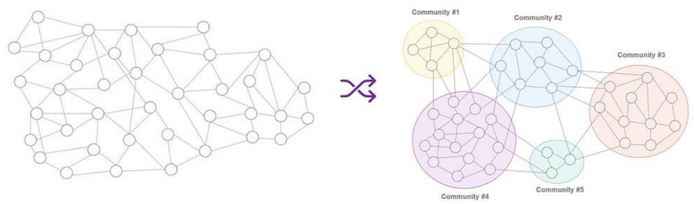
Abstract

Community detection plays an important role in extracting high-level information from a network. It mainly emphasizes the presence of dependencies inside a group and the absence of dependencies between groups. We review the concept of modularity, which is one common measure of the quality of a particular division of a network. Some modularity-based algorithms are developed to detect the community, for example, the spectral optimization method and the Louvain algorithm. We apply those methods to image segmentation.

Community Detection

For a given graph, or network, we want to put the nodes into clusters. This is known as community detection. We would like to evaluate the quality of a particular community of a network and provide appropriate methods and estimates to locate these communities.



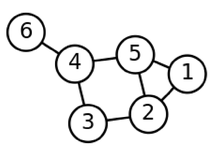
Graph Theory Background

Suppose Graph $G = (V, E, W)$ is a connected undirected graph with n vertices and m edges.

Vertex set V: $v_i \in V$

Edge set E: $e_{ij} = (v_i, v_j) \in E$

Weight set W: $w_{ij} \in W$



Degree $d_i = d(v_i)$: the number of edges connected to a vertex v_i

Weighted Degree $\delta_i = \delta(i) = \sum w_{ij}$

Unweighted Adjacency Matrix A:

$$a_{ij} = \begin{cases} 1, & \text{if } e_{ij} \in E \\ 0, & \text{otherwise} \end{cases}$$

Weighted Adjacency Matrix W:

$$W_{ij} = \begin{cases} w_{ij}, & \text{if } e_{ij} \in E \\ 0, & \text{otherwise} \end{cases}$$

Degree Matrix D:

$$D_{ij} = \begin{cases} \delta_i, & i = j \\ 0, & i \neq j \end{cases}$$

Laplacian Matrix L:

$$L = D - W$$

k-Partition of V: collection of subsets $P = \{S_1, \dots, S_k\}$ such that $\cup_i S_i = V$ and $S_i \cap S_j = \emptyset$ for $i \neq j$

For any $S \subseteq V$:

$|S|$: the cardinality of a set S

$\bar{S}$: the complement of $V \setminus S$

$vol(S) = \sum_{i \in S} d_i$: the **volume** of S

$vol(G) = \sum_{i \in S} d_i = 2m$: the volume of the whole graph

For any $S \subseteq V$, we let $\mathbf{1}_S$ be its **characteristic vector**, defined as

$$(\mathbf{1}_S)_i = \begin{cases} 1, & \text{if } i \in S \\ 0, & \text{otherwise} \end{cases}$$

For any subsets $S, T \subseteq V$, let $e(S,T)$ be the set of edges joining vertices in S with vertices in T, and let

$$e(S,T) = \mathbf{1}_S{}^T A \mathbf{1}_T = \sum_{i \in S} \sum_{j \in T} a_{ij}$$

Let $e_{in}(S) = e(S,S)$ and $e_{out} = e(S, \bar{S})$

Note: $vol(S) = e_{in}(S) + e_{out}(S)$

Configuration Random Model

To construct a random graph from the Configuration Random Model,

1. Select a degree sequence – the degree of each vertex is predefined. The sum of the degrees must be even (else, increase the degree of a randomly chosen node by 1).
2. Attach "stubs" to the nodes – if node i has degree d_i, attach d_i stubs.
3. Match the "stubs" to form edges – randomly match each stub to another stub and connect with an edge.

For large graphs, we do not need to worry about self-edges and multi-edges.

Probability that two vertices i, j are connected:

$$p_{ij} = \frac{d_i d_j}{2m - 1} \approx \frac{d_i d_j}{2m} = \frac{d_i d_j}{vol(G)} \text{ (for large m)}$$

Modularity [4][5]

Modularity is a measure of the structure of networks or graphs which measures the strength of division of a network into communities. Networks with high modularity have dense connections with nodes within their respective communities but sparse connections between nodes in different communities.

Modularity Matrix

The **modularity matrix** of undirected, unweighted graph $G = (V, E)$ is defined as:

$$M = A - \frac{1}{vol(G)} d d^{\mathrm{T}}$$

(i, j)–entry of M:

$$m_{ij} = a_{ij} - \frac{d_i d_j}{vol(G)}$$

- a_{ij} indicates the presence of an edge between nodes i and j
- $\frac{d_i d_j}{vol(G)}$ estimates the expected number of edges between vertices i and j if edges in the graph were placed with a uniform random distribution, according to the given degree sequence $d_1, \dots, d_n$

Then, the (i, j)–entry of M measures the disagreement between the expected and actual number of edges joining i and j.

A subset of vertices $S \subseteq V$ forms a community if the subgraph $G(S)$ contains a larger number of edges than expected.

Modularity of a Set

Given a graph $G = (V, E, W)$, consider a subset of vertices $S \subseteq V$. The **modularity** of S is defined as

$$Q(S) = e_{in}(S) - \frac{(vol(S))^2}{vol(G)}$$

We can also rewrite the modularity of S as

$$Q(S) = \mathbf{1}_S^T M \mathbf{1}_S$$

- $Q(S) = Q(\bar{S})$
- If $Q(S) > 0$, then the subgraph $G(S)$ contains "more edges" than expected from the reference model.

Modularity for Multi-Communities

The modularity of a partition $P = \{S_1, \dots, S_k\}$ of V is defined as

$$Q(P) = \frac{1}{vol(G)} \sum_{i=1}^{k} Q(S_i) = \frac{1}{vol(G)} \sum_{i=1}^{k} \mathbf{1}_{S_i}^T M \mathbf{1}_{S_i}$$

The normalization factor $\frac{1}{vol(G)}$ is purely conventional and for compatibility to settle the value of $Q(P)$ in a range independent of G.

Spectral Method [3][5]

Similar to the spectral method for graph partitioning based on the Laplacian matrix, we can follow the same idea to find two communities based on the modularity matrix:

1. Compute the eigenvector $\vec{u}_1$ corresponding to the largest positive eigenvalue λ_1.
2. Group the vertices according to the signs of the components of $\vec{u}_1$ to divide the graph into two groups.

For multiple communities, we only need to compute more corresponding eigenvectors and apply k-means method.

Louvain Algorithm [1]

Louvain Algorithm is an unsupervised method for extracting non-overlapping communities.

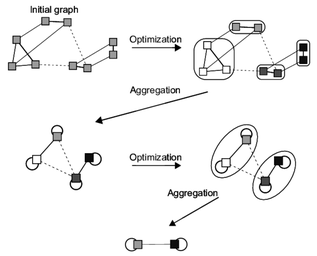
Phase 1: Modularity Greedy Optimization

1. Initialize partition: assign a different community to each node, so there are n initial communities.
2. For each node i, consider the neighboring nodes j of i and evaluate the gain of modularity by removing i from its community and placing it in the community of j.
3. Place node i in the community which has a local maxima of the gain of modularity. If the gain is not positive, keep i in its original community. End Phase 1 when no individual moves improve modularity.

Phase 2: Community Aggregation

1. Start with the new network of communities found during the first phase.
2. Get each element of the new aggregated weighted adjacency matrix $\vec{A}_{cd}$ between the new nodes c and d by adding the number of the links between nodes in the corresponding two communities c and d in the original graph.

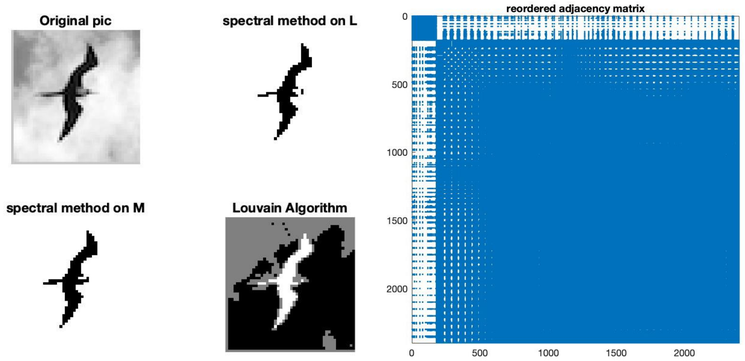
Reapply **Phase 1** to the resulting weighted network and iterate until no more changes and a maximum of modularity is attained.



Application: Image Segmentation

Compute weights: $w_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{\sigma^2}\right)$ with $\sigma = 2$, where x_i is the corresponding pixel.

Build unweighted adjacency matrix A: if $w_{ij} > 0.985, a_{ij} = 1$; Otherwise $a_{ij} = 0$



Conclusion

Community detection methods are used to discover strongly connected groupings of nodes (communities) within a network. We review the definition of modularity as a quality function used to optimize the detection of a community structure within a network. We then apply the spectral method and the Louvain method to image segmentation.

References

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